

Buoyancy Gravity Triadic Mode

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PAPER_963: Buoyancy Gravity Triadic Mode

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Source: triadic_solutions_next.py (BuoyancyGravityTriadic)

Calculator: BuoyancyGravityTriadicCalc (CP4 #547)

CVW: v2.0.0 compliant

Abstract

The Buoyancy Gravity Triadic mode evaluates $E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t) - \text{threshold}$. Positive E_{net} drives expansion (nebulae, HII regions); negative E_{net} drives erosion (filaments, cometary knots).

1. Net Energy

$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t) - \text{threshold}$

2. Regime Classification

E_{net}	Regime	Astrophysical Example
> 0.1	Expansion	Nebulae, HII regions
< -0.1	Erosion	Filaments, pillars
$[-0.1, 0.1]$	Neutral	Transition zones

References

- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- PAPER_961 — Compressed Gravity Triadic
- PAPER_962 — Resonant Gravity Triadic
- PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch of same triadic

PAPER_962	Resonant branch
PAPER_966	Unified solver combining all three
PAPER_954	$E(t)$ sign-flip from buoyancy dynamics

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_U B_i$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy coupling
E_{net}	sign	—	$+$: expansion, $-$: erosion Phase indicator

SM Anchor — CVW v2.0.0

Observable UQFF Prediction Status
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E_{net} sign-flip Expansion $\rightarrow$ erosion phase transition Derived
F_{UBi} buoyancy force Acts outward against g_{comp} Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:** Buoyancy Gravity (F_{UBi} Net Energy Balance)**

§A.2 Lagrangian Density $\mathcal{L}_{\text{buoy}} = F_{\text{UBi}}(r) \cdot r - \int_0^r g_{\text{comp}}(r') \cdot dr'$

§A.3 Euler-Lagrange Equation of Motion $\boxed{E_{\text{net}}(r) = F_{\text{UBi}}(r) \cdot r - \int_0^r g_{\text{comp}}(r') \cdot dr'}$, $\text{sign}(E_{\text{net}})$: $+\text{expand}$, $-\text{erode}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ buoyancy force F_{UBi} $\rightarrow$ net energy balance $\rightarrow$ expansion/erosion phase $\rightarrow$ triadic branch 3/3

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS F_{UBi} profile traces VDS outward pressure gradient.

§B.2 DVP Buoyancy vortex dipole: inward gravity vs. outward F_{UBi} .

§B.3 BSH E_{net} saturates at stellar surface (BSH envelope crossing).

§B.4 Production-Scale Consistency

Metric Value Status
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β_{i} 0.603 Confirmed
$[\text{SSq}]$ 0.57 Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper Title
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PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

6 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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